

MS20 MKII Multimode Resonant Filter for Eurorack Synthesizers

By baritonomarchetto in Circuits > Audio  976  1



Filters are one of the most interesting components of a synthesizer. They assolve a fundamental role in defining the "character" of a musical device and very likely are the most important weapon a sound engineer has in its hands to shape the "perfect" sound.

One of my favourite filter is the one Korg developed for its MS20 synthesizer. In particular, I must admit that I have a soft spot for the so called "Mark II" filter because less aggressive then more suited to my musical tastes than the Korg35 custom chip (*shame on meeee*).

In this Instructable I will show you my new Eurorack filter module based on MS20 MKII filter schematics.

I will introduce you to it's building blocks, give you a *tour* on how it has been cloned/reproduced by talented DIY tinkerers with components they had by hand, and I will show you how I managed to realize my new version of the full Eurorack filter module.

As always, I will share with you Gerber files to have module's PCBs realized and speed up the building process ;)

There we go!

Supplies

The module is made of three PCBs: a front panel, a front board and a main board.

Main board comes in two flavours: full through-hole (THT) components and mixed THT and surface mount (SMD) components

I generally prefer to use THT components because they are easier to solder, and I especially like the possibility to swap them with other equivalents to ease the differences. In this very special case, the OTA (LM13700) is now difficult-to-impossible to find in DIP format. There are (most often working) clones one can buy, but if you want a legit chip you are forced to use LM13700 in SMD format (SOIC).

This is why I designed a main board with some SMD components: the OTA, the two op-amps and the transistor pair.

In the following the Bill Of Materials for main boards and front board. The front panel has no components.

Where components are different from SMD and THT versions, they are reported in parentheses.

Front Board

Resistors, trimmers, potentiometers

3x 100K ohm potentiometer (WH148)

1x 10K ohm trimmer (B25P)

1x 1k ohm resistor

1x 2.2K ohm resistor

2X 100K ohm resistor

1x 220k ohm resistor

Others

2x 8 pin header, MALE, 0.1" spacing

4x PJ301M mono female jack connector

1x MTS202 DPDT switch

Main Board

Transistors, amplifiers, diodes

1x BC857BS PNP general-purpose double transistor, SOT363 (only SMD version)

2X BC557 PNP general-purpose transistor (only THT version)

1x LM13700M LM13700M DUAL-OTA transitor (SOIC16/DIP16)

2x TL072D generic OP-AMP (SOIC8/DIP8)

2x LED (white), 3.5 mm spacing

Resistors, trimmers

1x 1K ohm trimmer (B25P)

4x 220 ohm resistor

1x 1.5k ohm resistor

2x 4.7K ohm resistor

6x 10K ohm resistor

1x 47K ohm resistor

1x 470K ohm resistor

Capacitors

2x 1nF non polarized capacitor

1x 4.7nF non polarized capacitor

5x 100nF non polarized capacitor

1x 470nF non polarized capacitor

2x 10uF electrolytic capacitor

Others

1x IDE connector, 5x2 pin

2x 8 pin header, FEMALE, 0.1" spacing

Step 1: The Korg MS20: Why?



Short answer

Because it's beautiful (attached pictures are of my beloved 1980 unit!)

Long answer

The Korg MS-20 is one of the most iconic monophonic synthesizers ever produced. Initially released in 1978, the original MS-20 features a distinctive, dual high pass/low pass resonant filter design, that mostly contribute to its aggressive and raw sound.

Due to its enduring popularity, Korg has released several new versions of the MS-20 over the years, starting from the "noisy" Korg MS-20 Mini in 2013, to various clones by third-party manufacturers like the Behringer K2, now at it's second attempt.

The MS-20's semi-modular architecture has always fascinated me, with it's 1/4" jack connectors and it's "experimental" look. Semi-modular is nowadays a standard adopted by most synth Companies, but at the time the feature was intentionally left out on cheap instruments.

Sound-wise, there's nothing as gorgeous as MS20 filters section, which continues to inspire synthesists, producers and, you can bet it, tinkerers. The filter's ability to self-oscillate and produce rich harmonic distortion makes it particularly popular for creating resonant sweeps and biting, aggressive bass tones but also re-sculpture whole tracks and samples (1996 *Daft Punk anyone?* :)).

This being said: **Could you imagine something more attractive than enclosing the MS20 essence in a 6HP eurorack module? ;)**

Step 2: The Korg MS20 MKII Filter

The MS-20 filter went through two major revisions in its design.

The Korg35 filter design (early models, pre-1980) use a custom chip which provides a more aggressive resonance, but is known for instability at high resonance settings.

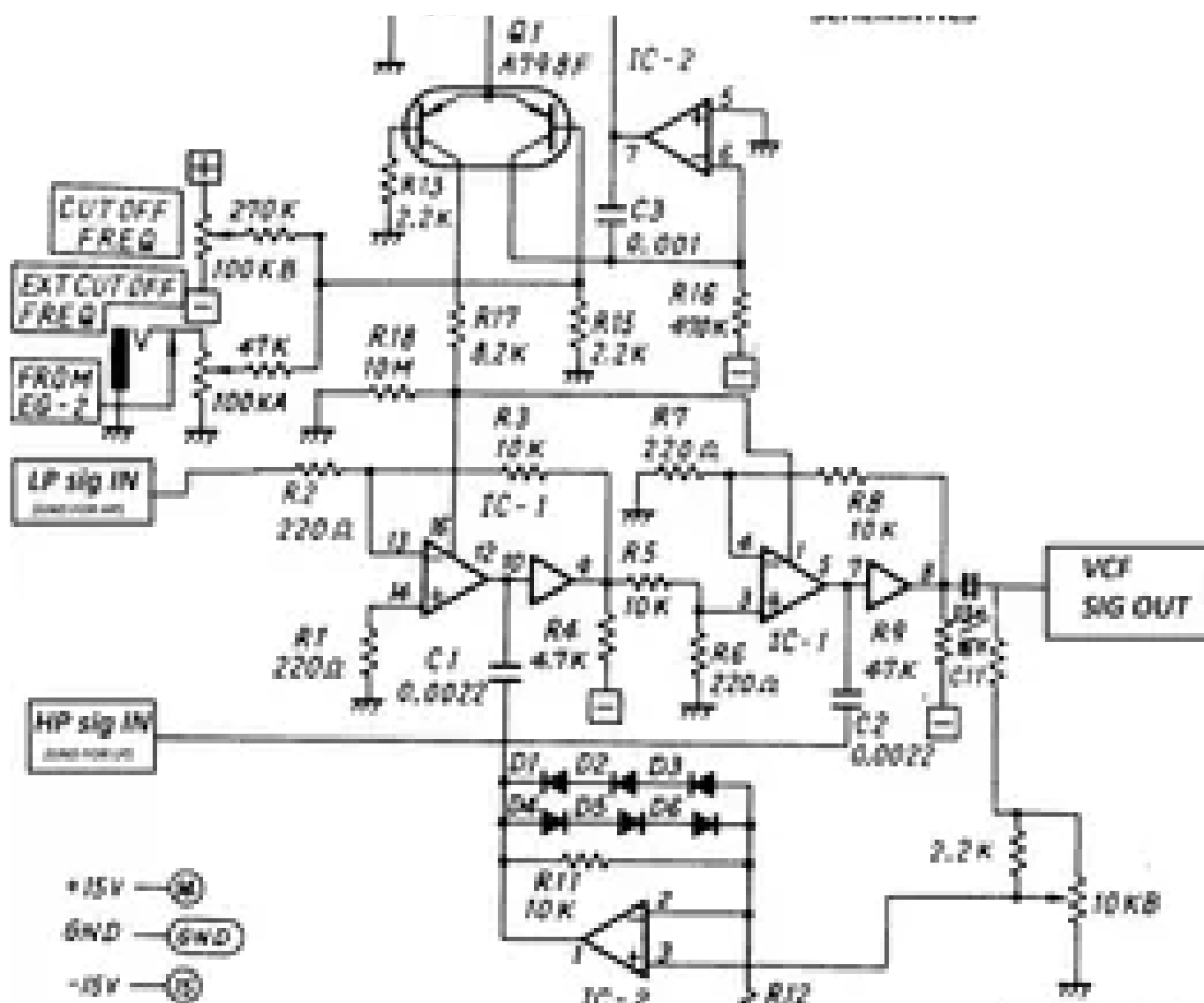
Later models adopted an Operational Transconductance Amplifier (OTA) filter. This has a smoother and more predictable response while retaining the character of the original. The resonance feedback mechanism remains similar, but with improved predictability and control over self-oscillation.

The MS-20 MKII filter circuit consists of three key elements:

- **Operational Transconductance Amplifiers.** The core filtering is achieved through LM13600 OTA. These amplifiers dynamically adjust their transconductance based on an input control current, allowing for precise cutoff frequency modulation.
- **Resonance feedback loop.** A portion of the filter's output is fed back into the input to reinforce specific frequencies, creating resonance. This is controlled via a variable resistor that adjusts the amount of feedback applied.
- **Voltage-Control tracking circuit.** Enables dynamic control over the filter's cutoff frequency based on an external CV or the front panel knobs.

In the following Step I will try to explain with further details the circuit design of the MKII filter.

Step 3: A Little More in Deep With the Filter Circuit



Attached is a (slightly) modified schematic of the MS20 MKII OTA filter section. The circuit adopted in this project (*and any other project you will find online at the time of this writing*) is based on this "later" version.

OTAs are a very special class of amplifiers. In a nutshell, they are amps with current controllable gain. This makes it possible to obtain various interesting circuits for our modules, where voltage control is a must-have.

As we can learn by reading [this great writing from Tom Wiltshire](#) @electricdruoid.net or takeing a look at [LM13700 datasheet](#), even if voltages up to 5V could hit OTAs inputs, OTAs can only cope with low input levels before "excessive" distortion. This is why a voltage divider is in the need on the input hit by the signal (not shown in the attached circuit diagram, but you can see a small value resistor to ground just before it's inverted input).

The output from the chip is a current, not a voltage, so out of each OTA in the schematics you see a more classic op-amp acting as current-to-voltage converter.

In the upper part of the circuit diagram we can see the external voltage to OTAs input bias current converter. The current to control the OTA's gain has also to be very low: go higher than 2 mA and the OTA is blown. This circuit is in charge to sum the input voltages coming from the cutoff potentiometer and external modulator and convert them to currents the OTA can handle without smoking out.

The lower part of the circuit is the feedback circuit. A fraction of the output level (determined by the resonance potentiometer) feeds back into the circuit. The schematic also shows a practical way to turn a linear pot into a (inverse) logarithmic pot.

The feedback signal goes through an op-amp with *back-to-back* diodes. The effect of these diodes is to deliberately distort the signal being filtered. If the loop voltage is sufficiently small such that the diodes don't conduct, the op-amp operates as a normal non-inverting op amp; once the loop voltage is such that the diodes conduct, the op-amp works as a (unity-gain) voltage follower, with a DC-offset given by the diode drop (a more in-depth description in [Tim Stinchcombe study](#)).

The circuit can work as low pass or high pass filter. The two operating modes are set by changing the position where the input signal enters the circuit.

Notice that the low pass circuit has two nodes, so it's of the second order; the high pass circuit has a single node, so it's of the first order.

What if you place two of these filters in series? Well, you increase the number of nodes and, in the end, the response of the filter. Nice :) .

Step 4: Modern MS20 Clones Filter Circuit Design

There are various projects developed around the MKII filter out there. Just to cite a few, take a look at [René Schmitz](#) or [Kassutronics](#) work. Kassutronics one is particularly interesting because of the implementation of a blending function between the two filter modes.

Original MKII filter's active components are LM13600 Dual Operational Transconductance Amplifiers (OTA), some generic amplifiers in buffer configuration at the LM13600 outputs and a JRC4558 dual op-amp in the feedback circuit. A dual PNP transistor in single package (A798F) is at the core of the circuit taking care of CV tracking.

These components are mostly obsolete, but there are very good replacements available.

The LM13700 OTA is a direct replacement for 13600. No need to look further :)

High performane amps for audio applications are nowadays common (e.g. the NE5532), but you can bet even "cheaper" op-amps (like the TL072) are likely better suited for the task than those available in the '80's (*yes, you want to use legit integrated circuits for such a delicate task and, yeees, these are great years for being a tinkerer!!*).

The A798F dual transistor in the exponential converter circuit is commonly replaced by two BC557. Matching these is a good practice for better tracking, but not mandatory (it's not a VCO, in the end).

Step 5: My Project Circuit Design

This project's circuit design doesn't add too much to those cited in the previous step.

I used TL072 for audio and feedback paths. I made some testing in the previous filter versions by using TL072 in place of NE5532 (being regarded as more preferable op-amp for audio) and, honestly, could not hear any difference.

The input level to the first OTAs channel is commonly kept relatively high (remember the "keep signal low or distortion will trigger" thing?) being in the 400 mA ballpark with an input level of 10Vpp. This is generally adopted online, and I could argue it's because some distortion at this stage of the filter contributes to it's character.

My 2 cents here is the introduction of a trimmer to attenuate the signal hitting the first OTA. The trimmer is accessible from the front panel and is dimensioned so that you can load the OTAs input well above it's confort zone when fully counter clockwise and gradually attenuate it turning clockwise.

In the SMD version of the main board, I have used a dual PNP transistor in a SMD package (BC857BS) for the CV circuit. Being in the same device makes the two transistors work the same at any temperature and operating condition. This makes them "almost matched" and should allow for better voltage tracking.

Perfect tracking is not that important in a filter with respect to a voltage controlled oscillator, where a reliable pitch tracking is mandatory, but, hey: we are trying to do our best here :)

Step 6: Module Design



The module is designed for compatibility with standard Eurorack synthesizer formats, ensuring ease of assembly and integration.

I spent due efforts to keep the module width as small as possible, ending at 6 HP.

The module features two CV inputs, one direct and the other with control over CV level. CV level is a simple potentiometer in voltage divider configuration, so nothing fancy.

This design operates either as low-pass or high-pass filter. Changing operation mode is as simple as flipping a switch from one position to another.

This configuration flexibility allows for diverse sound-shaping possibilities within a modular synthesizer setup. You can use a single filter and have a two poles filtering, or increase the number of filters in series thus increasing the number of poles. If you want to reproduce the MS20 filter configuration, place two of these in series, the first one in HP configuration, the second one in LP config and there you are :) .

The module is made of three boards intended to be stacked one over the other. The first one is the front panel. This has no traces and is intended for mechanical installation on your eurorack case. It is preferred to be realized in alluminum alloy.

The other two PCBs are the front board and the main board. Front board hosts jacks, mode switch and control potentiometers. The main board hosts most of the filter circuitry.

The module is intended to work on +/-12V, but should work on +/-15V without problems.

As already said, I have layed down not one but two versions of the main board for this filter. The first one is made of THT components only. It's easier to build, but you could have some trouble finding legit components for it. In the moment I am writing, *four-out-of-four* main sellers don't have the LM13700 in DIP16 package in stock (as far as I know Texas Instruments has discontinued the production of the DIP version some year ago, while the SOIC is still in production).

This is why I also layed down a version with some SMDs. This version could be more complex to assemble by yourself, but at least you can find legit SOIC16 LM13700!

Step 7: SMD Soldering Hints

Soldering SMD components, especially those with small pitch, calls for some strategy to successfully solder it in place with a common stylus soldering station.

Some suggestion follow.

General Procedure:

- **solder SMD components *first***. You want space around your solder tip and you don't want to trash other components in the case of problems.
- use a **thin iron** to solder the chip.
- clean the iron frequently. Solder wire has a flux core that degrade after a while. The solder becomes "sticky" and difficult to handle.
- use a magnifying glass to zoom the soldering area (even better would be one of those digital microscopes they sell nowadays, but it's not everybody priority I know).
- use soldering flux. My personal favourite are siringes containing dense flux.
- The iron temperature is very important: it should be high enough to fast melt the solder, but not too high or it will kill the IC. I set mine at temperatures of 370-380°C.

Soldering Method:

- put flux in the IC's pads whole area
- melt a little amount of solder on one PCB's IC pad. Choose one of those pads with a side free from other pads (pin 1, in example). You want to see a certain amount of solder over the pad, but no shorts between adjacent pads.

- place the SMD IC over it with the right orientation and apply soldering flux.

Be 100% sure the IC has the right orientation!!

- remelt the pad-solder with a clean iron tip while applying a very small pressure on top of the IC.

When the first IC leg is firmly soldered, check if all legs are well aligned to their respective pads. If yes, use some flux and fresh solder wire to solder all the other legs to pads, starting from those on the opposite side to the one firstly soldered.

It can happen that two (or more) legs are shorted with solder. If this happen, clean the soldering iron and pass it through the shorted legs side moving it from up to down or viceversa. The excess solder will stick to the iron and the short will go.

Alternatively...

If you don't feel confident soldering small SMD components, be aware that most PCBs manufacturing Companies offer SMD assembly service you can take advantage from.

Main board PCB you see in this Instructable pictures has been partially populated by JLCPCB SMD assembly service. Very good results indeed!

They have legit LM13700 in stock, so it's a good way to catch two birds with a single net ;)

Please notice that only SMD components will be assembled; THT components will need to be soldered.

Step 8: Acknowledgments



Many thanks to JLCPCB for sponsoring the manufacturing of PCBs and the assembly of main board's SMDs for this module.

Without their contribution this project would have never reached it's actual level of development.

JLCPCB is a high-tech manufacturer specialized in the production of high-reliable and cost-effective PCBs. They offer a flexible PCB assembly service with a huge library of more than 600.000 components in stock.

3D printing is part of their portfolio of services so one could create a full finished product, all in one place!

By registering at JLCPCB site via [THIS LINK](#) (affiliated link) you will receive a series of coupons for your orders. Registering costs nothing, so it could be the right opportunity to give their service a due try ;)

All Gerber files and sketches I realized for this project are stored >>HERE<< (Github).